

Five test datasets pushed. Here's exactly how to use each one to test every new tool:

Files in **samples/** — download from GitHub or drag straight into GeoSpaX

File	Features	Purpose
butterfly_presences.geojson	40 pts	Presence layer for all 3 SDMs
vine_host_plants.geojson	53 pts	Host plant layer — Hotspot Grid + WLC
habitat_survey_grid.geojson	90 pts	Environmental variable grid — all SDMs
forest_patches.geojson	35 pts	WLC criterion + constraint layer
stream_water_sources.geojson	28 pts	Additional WLC/env layer

Test workflow per tool

Conservation Hotspot Grid

1. Load: butterflies + host plants + habitat grid
2. Analysis → Hotspot Grid → Hex, 0.15°, Count
3. Assign weights: butterflies=3, host plants=2, habitat=1
4. Expected: two hot cells matching the lowland (~146.15°E, 6.38°S) and montane (~146.52°E, 6.81°S) colonies

Suitability WLC

1. Load: butterflies + host plants + forest patches
2. Analysis → Suitability WLC → set forest patches as constraint layer
3. Use **density** (host plants) and **count** (butterflies) as fields
4. Expected: green patches in butterfly zones, forced to zero in high-disturbance forest patches

Bioclim SDM

1. Load: butterfly presences + habitat grid
2. SDM → Presence = butterflies, Env layer = habitat grid, Field = **elevation_m**
3. Add second env layer = habitat grid, Field = **rainfall_mm**

4. Model = Bioclim, Resolution = 0.08°
5. Expected: green band at 200–600m (lowland colony) + 900–1400m (montane colony)

Mahalanobis SDM

- Same setup as Bioclim but Model = Mahalanobis
- Expected: elliptical suitability zones around the env centroid of presences; more gradual falloff than Bioclim

MaxEnt (server)

- Same setup, Model = MaxEnt
- Calls `/api/sdm` on Vercel — requires the live deployment to be running
- Expected: probability surface similar to Mahalanobis but with regularised logistic-regression boundary